



Wave component in the Green function for diffraction radiation of regular water waves

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ABSTRACT

The Green function for diffraction radiation of regular waves in deep water is considered. The Green function G and its gradient ∇G involve non-oscillatory local-flow components L and ∇L , for which simple global approximations valid within the entire flow region exist, and wave components W and ∇W . The waves W and ∇W in this basic decomposition involve the intrinsic Fortran Bessel functions $J_0(h)$ and $J_1(h)$, where h denotes the horizontal distance between the source and flow-field points in the Green function, and the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ for which complementary approximations valid within the nearfield or farfield ranges $0 \leq h \leq 3$ or $3 < h$ exist. These complementary approximations to $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ defeat the global nature of the approximations to the local-flow components L and ∇L . This issue, however, is readily remedied if practical approximations, given by Aarts and Janssen in 2016, that relate the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ to the Bessel functions $J_0(h)$ and $J_1(h)$ are used. The resulting approximations to the waves W and ∇W given here, and the global approximations to the local-flow components L and ∇L given previously, yield practical and particularly simple approximations to G and ∇G that are valid within the entire flow region, can be evaluated simply and efficiently, and are sufficiently accurate for practical applications.

1. Introduction

Diffraction radiation of time-harmonic water waves by an offshore structure within the framework of linear potential flow theory and the Green function method is widely used to predict added-mass and wave-damping coefficients, motions, and wave loads. Wave diffraction radiation by a ship that travels at low forward speed in regular waves is also widely analyzed via the zero-speed Green function at the encounter frequency. This Green function, related to the potential of the flow created by a pulsating point source as is well known, is an essential element of the theory of wave diffraction radiation. Accordingly, the Green function has been widely studied in a broad literature, especially for the simplest case of deep water that is considered here. A brief review of this literature can be found in [1,2].

The Green function G can be expressed as the sum of the fundamental free-space singularity and a flow component that accounts for free-surface effects. This free-surface component is formally decomposed into a wave component W that represents the waves radiated by the pulsating source and a non-oscillatory local-flow component L in [3], where several integral representations and complementary

analytical approximations (nearfield series, farfield asymptotic expansions, and one-dimensional Taylor series) to G and ∇G are given. These alternative approximations are the foundation of the method for computing G and ∇G given in [4]. Alternative mathematical representations and computational methods (notably polynomial approximations in complementary contiguous regions, and table interpolation associated with function and coordinate transformations) for approximating G and ∇G have been given in the literature, reviewed in [1,2] as was already noted.

In particular, the method for computing the local-flow components L and ∇L given in [4] requires subdivision of the flow region into five contiguous regions of space within which different analytical approximations are used. Other methods given in the literature are similarly based on complementary polynomial approximations in contiguous subdomains.

A significantly different approach is adopted in [1] where analytical approximations valid within the entire flow domain, i.e. global approximations, are given for the local-flow components L and ∇L . These global analytical approximations to L and ∇L are shown in [2] to be sufficiently accurate for practical applications.

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Simple practical global approximations (valid within the entire flow region) to the exact expressions for W and ∇W , used in [1,2], are given here. These approximations to W and ∇W are based on the approximations, given in [5], that express the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ in terms of the Bessel functions $J_0(h)$ and $J_1(h)$.

The global analytical approximations to the wave components W and W_h , together with the global analytical approximations to the local-flow components L and L_h given in [1,2], provide practical global analytical approximations to the Green function G and its gradient ∇G for diffraction radiation of regular water waves. These global analytical approximations are particularly simple, well suited for parallel computations, and are shown to be sufficiently accurate for practical applications.

2. The Green function and its gradient

A Cartesian system of coordinates $\mathbf{X} = (X, Y, Z)$ is used. The Z axis is vertical and points upward, and the undisturbed free surface is taken as the plane $Z = 0$. Diffraction radiation of time harmonic waves with radian frequency ω and wavelength $\lambda = 2\pi g/\omega^2$, where g denotes the gravitational acceleration, is considered. Nondimensional coordinates $\mathbf{x} = (x, y, z) = \mathbf{X}\omega^2/g$ are defined.

The Green function $G(\mathbf{x}, \tilde{\mathbf{x}})$ is the spatial component of a non-dimensional velocity potential $\text{Re}[G(\mathbf{x}, \tilde{\mathbf{x}})e^{-i\omega T}]$ where T denotes time, and corresponds to the potential of the flow created at the point $\mathbf{x} = (x, y, z \leq 0)$ by a pulsating source located at the point $\tilde{\mathbf{x}} \equiv (\tilde{x}, \tilde{y}, \tilde{z} < 0)$, or flux through the free surface at the point $\tilde{\mathbf{x}} \equiv (\tilde{x}, \tilde{y}, \tilde{z} = 0)$.

The nondimensional distances between the flow-field point $\mathbf{x}$ and the source point $\tilde{\mathbf{x}}$ or its mirror image $\tilde{\mathbf{x}}_1 \equiv (\tilde{x}, \tilde{y}, -\tilde{z})$ with respect to the undisturbed free-surface plane $z = 0$ are denoted as r and d , and are given by

$$r \equiv \sqrt{h^2 + (z - \tilde{z})^2} \quad \text{and} \quad d \equiv \sqrt{h^2 + v^2} \quad \text{where} \quad (1a)$$

$$0 \leq h \equiv \sqrt{(x - \tilde{x})^2 + (y - \tilde{y})^2} \quad \text{and} \quad v \equiv z + \tilde{z} \leq 0 \quad (1b)$$

are the horizontal and vertical components of the distance d between the points $\mathbf{x}$ and $\tilde{\mathbf{x}}_1$.

The Green function G and its gradient $\nabla G \equiv (G_x, G_y, G_z)$ are expressed in [1–3] as

$$4\pi G = -1/r + L + W \quad (2a)$$

$$4\pi G_z = (z - \tilde{z})/r^3 + v/d^3 - 1/d + L + W \quad (2b)$$

$$4\pi G_h = h/r^3 + L_h + W_h \quad (2c)$$

$$G_x = G_h(x - \tilde{x})/h \quad \text{and} \quad G_y = G_h(y - \tilde{y})/h \quad (2d)$$

where the term $-1/r$ and its derivatives correspond to the free-space Green function, and the functions $L(h, v)$, $L_h(h, v)$, $W(h, v)$ and $W_h(h, v)$ account for free-surface effects. The terms L and L_h correspond to a non-oscillatory local flow, and the terms W and W_h represent circular surface waves radiated by the source.

3. Local-flow components L and L_h

Simple analytical approximations to the local-flow terms L and L_h in (2a)–(2c) are given in [1]. These approximations are valid within the entire flow domain ($0 \leq h, v \leq 0$), i.e. are global approximations, unlike the alternative approximations in complementary contiguous regions given in the literature. The approximations to the local-flow components L and L_h given in [1] only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments, and provide a particularly simple practical method for numerically evaluating the local-flow components L and ∇L in G and ∇G . These local-flow approximations are shown in [2] to be sufficiently accurate for practical applications.

4. Wave components W and W_h

The wave component W in (2a) and (2b) and its derivative W_h in (2c) are expressed in [1–3] as

$$W(h, v) = 2\pi [\tilde{H}_0(h) - iJ_0(h)]e^v \quad (3a)$$

$$W_h(h, v) = 2\pi [2/\pi - \tilde{H}_1(h) + iJ_1(h)]e^v \quad (3b)$$

where $J_0(h)$, $J_1(h)$ and $\tilde{H}_0(h)$, $\tilde{H}_1(h)$ denote the common Bessel or Struve functions. The Bessel functions $J_0(h)$ and $J_1(h)$ are intrinsic Fortran functions and are then readily evaluated. As is noted in [1], several approximations for the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ can be found in the literature; E.g., complementary approximations valid in the regions $0 \leq h \leq 3$ or $3 < h$ are given in [6].

These complementary approximations in contiguous regions defeat the global nature of the approximations to the local-flow components L and ∇L given in [1]. This drawback of the exact expressions (3) for the wave functions W and ∇W is remedied if one uses the analytical approximations for $\tilde{H}_0$ and $\tilde{H}_1$ given in [5]. Specifically, these approximations are

$$\tilde{H}_0(h) \approx J_1(h) + A_0(1 - \cos h)/h - B(\sin h - h \cos h)/h^2 - C(h^* - \sin h^*)/h^2 \quad (4a)$$

$$2/\pi - \tilde{H}_1(h) \approx J_0(h) - A_1(\sin h)/h - B(1 - \cos h)/h^2 + C(1 - \cos h^*)/h^2 \quad (4b)$$

where A_0, A_1, B, C and h^* are defined as

$$A_0 \equiv 1.134817700; \quad A_1 \equiv 0.0404983827 \quad (5a)$$

$$B \equiv 1.0943193181; \quad C \equiv 0.5752390840 \quad (5b)$$

$$h^* \equiv 0.8830472903h \quad (5c)$$

The approximate relations (4) express the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in terms of trigonometric functions and the Bessel functions J_1 and J_0 . The approximate relations (4)–(5) are shown in [5] to be sufficiently accurate for practical purposes.

5. Illustrative numerical applications

For purposes of illustration and validation, linear and mean drift wave loads are computed here for a hemisphere and a freely floating FPSO, as in [2]. The local-flow components L and L_h in the Green function G and its gradient ∇G are evaluated via the global approximations given in [1]. These approximations are shown in [2] to be sufficiently accurate for practical purposes. The Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$ in the wave components W and ∇W are evaluated via Newman's approximations [6] or via the global approximations (4)–(5).

Wave radiation by a floating hemisphere, for which an analytical solution exists [7], is considered first. The hemisphere is discretized as in Fig. 1. The added-mass coefficients a_{11} and a_{33} for surge and heave are dimensional with respect to $2\pi\rho R^3/3$, and the corresponding wave-damping coefficients b_{11} and b_{33} are dimensional with respect to $2\pi\rho\omega R^3/3$, where ρ denotes the water density, R is the radius of the hemisphere, and ω is the circular frequency of the time-harmonic oscillatory motions of the hemisphere.

The added-mass and wave-damping coefficients a_{11} , a_{33} and b_{11} , b_{33} are depicted in Fig. 2 for adimensional wavenumbers $0.1 \leq k_0 R \leq 10$ where $k_0 \equiv \omega^2/g$. Fig. 2 shows that the numerical predictions in which the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in expressions (3) for the wave components W and W_h are evaluated via Newman's approximations given in [6] or via the global approximations (4)–(5) cannot be distinguished, and are in excellent agreement with Hulme's analytical results [7].

The second-order mean drift forces and moment related to the quadratic terms in Bernoulli's equation are now evaluated for the hemisphere already considered in Fig. 2 and for a FPSO, discretized as in Fig. 3. The incident waves acting on the hemisphere or the FPSO are determined by the potential

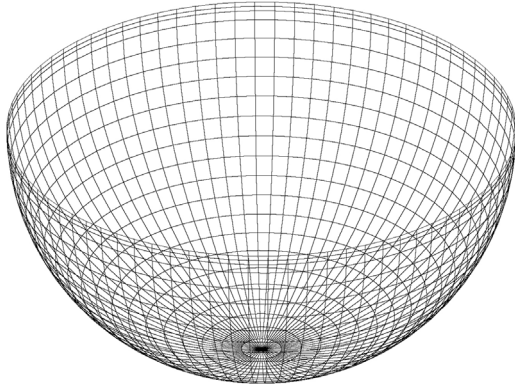


Fig. 1. Mesh used to discretize a hemisphere via 1761 nodes and 1720 quadrilateral elements.

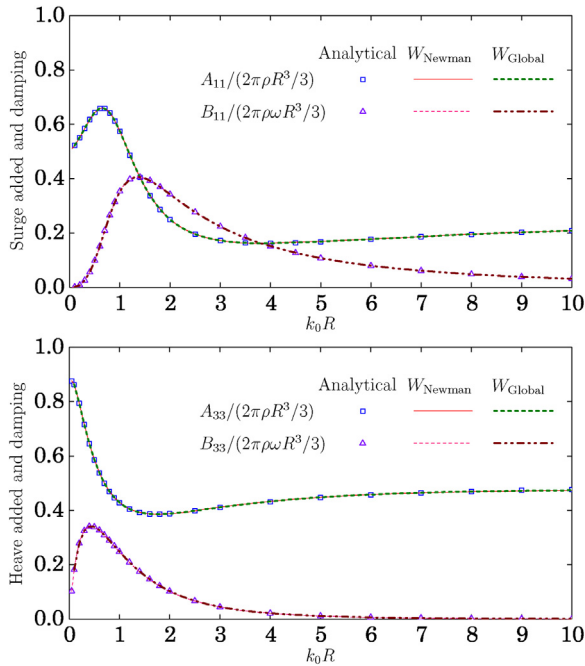


Fig. 2. Adimensional added-mass and wave-damping coefficients a_{11} and b_{11} for surge (top), and corresponding coefficients a_{33} and b_{33} for heave (bottom), predicted by Hulme's analytical solution [7], or a panel method where the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in the wave components W and W_h are evaluated via Newman's approximations [6] or via the global approximations (4)–(5).

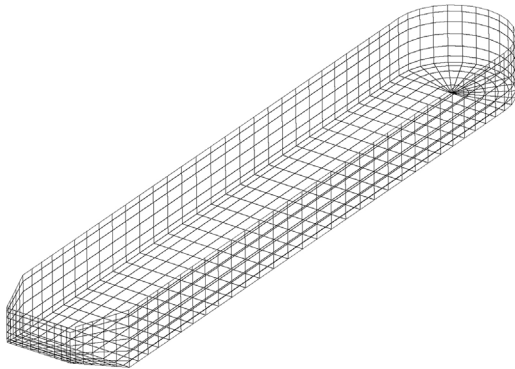


Fig. 3. Mesh used to represent the hull surface of the FPSO considered in the study. 503 quadrilateral panels are used to approximate half of the hull surface.

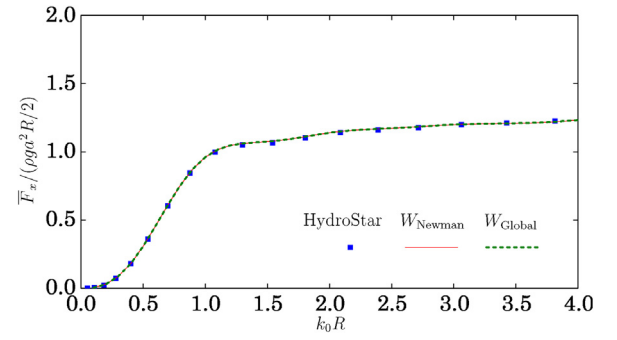


Fig. 4. Mean drift force $\bar{F}_x$ acting on a fixed hemisphere in incoming regular waves of amplitude a at an incidence angle $\beta = 0^\circ$, where $\bar{F}_x$ is predicted by HydroStar or by a panel method in which the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in the wave components W and W_h are evaluated via Newman's approximations [6] or via the global approximations (4)–(5).

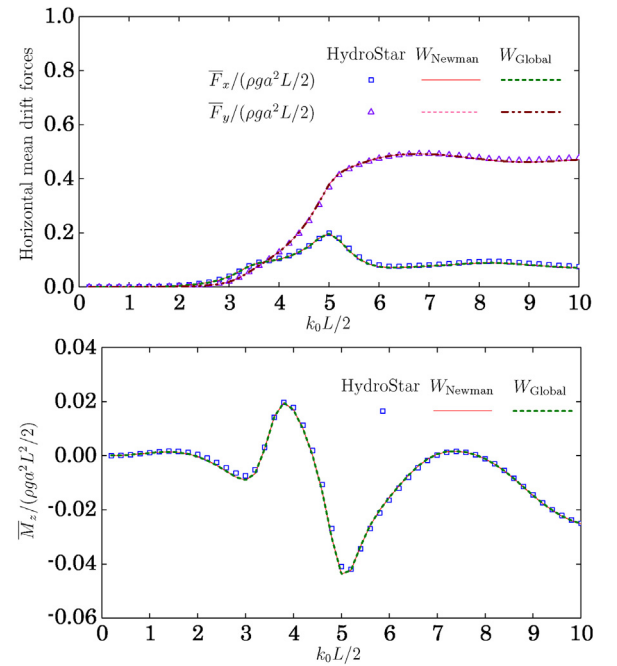


Fig. 5. Mean drift forces $\bar{F}_x$ and $\bar{F}_y$ and moment $\bar{M}_z$ acting on a freely floating FPSO of length L in incoming waves of amplitude a at an incidence angle $\beta = 45^\circ$, where $\bar{F}_x$, $\bar{F}_y$ and $\bar{M}_z$ are predicted by HydroStar or a panel method in which the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in the wave components W and W_h are evaluated via Newman's approximations or the global approximations (4)–(5).

$$\phi^I = (iag/\omega) e^{k_0[z + i(x\cos\beta + y\sin\beta)]} \quad (6)$$

where a , ω and β denote the wave amplitude, frequency and incidence angle.

Fig. 4 depicts the mean drift force $\bar{F}_x/(\rho g a^2 R/2)$ acting on the hemisphere for adimensional wavenumbers $0 < k_0 R \leq 4$. Similarly, Fig. 5 depicts the mean drift forces $(\bar{F}_x, \bar{F}_y)/(\rho g a^2 L/2)$ and the mean drift moment $\bar{M}_z/(\rho g a^2 L^2/2)$ acting on the FPSO in oblique waves at an incidence angle $\beta = 45^\circ$ for adimensional wavenumbers $0.1 \leq k_0 L/2 \leq 10$, where L is the length of the FPSO.

Figs. 4 and 5 show that the predictions in which the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ in expressions (3) for the wave components W and W_h are evaluated via Newman's approximations given in [6] or via the global approximations (4)–(5) cannot be distinguished, and are in excellent agreement with the predictions given by the Bureau Veritas (BV) software HydroStar, in which G and ∇G are evaluated with high accuracy. The very small differences between the HydroStar predictions and the predictions associated with the approximations W_{Newman} and W_{Global} to

W and W_h mainly stem from the fact that combined source and dipole distributions are adopted in the computations considered here, whereas only sources are used in HydroStar.

The numerical results depicted in Figs. 2, 4 and 5 show that the global approximations (3)–(5) to the wave components W and W_h in the Green function and its gradient are sufficiently accurate for practical applications.

The computing times required to evaluate the wave components W and W_h via Newman's approximations [6] or via the global approximations (3)–(5) are now compared. These alternative approximations to W and W_h are implemented in FORTRAN90 and compiled with Intel Fortran. The computations are performed on a laptop with processor of Intel(R) Core(TM) i7-7700HQ @ 2.8 GHz.

The CPU times for 10^7 evaluations of the global approximations to the local-flow functions L and L_h given in [1] and the global approximations (3)–(5) to the wave functions W and W_h given here are about 1.25 or 1.7 seconds, respectively, and thus represent about 42% or 58% of the CPU time required to evaluate both the local-flow and the wave components, i.e. the Green function and its gradient. The larger computing time required for the wave functions W and W_h is due to the fact that expressions (3)–(5) involve the exponential function e^v , the two Bessel functions $J_0(h)$ and $J_1(h)$, and the four trigonometric functions $\cosh$, $\sinh$, $\cosh^*$, $\sinh^*$, whereas the global approximations for the local-flow functions L and L_h given in [1,2] involve one logarithmic function, two exponential functions and polynomials.

If Newman's approximations to the Struve functions $\tilde{H}_0$ and $\tilde{H}_1$ are used in (3), the CPU times for 10^7 evaluations of the wave components W and W_h are about 1.4 or 3.3 s for $h \leq 3$ or $3 < h$, whereas the CPU time is about 1.7 s for every values of h and v if the global approximations (4)–(5) are used in (3) as was already noted. Thus, numerical evaluation of W and W_h via Newman's approximations for the Struve functions is about 18% faster or 94% slower for $h \leq 3$ or $3 < h$ than if the global approximations (3)–(5) are used. These tests are based on single-core computations and involve an 'if statement' to determine if $h \leq 3$ or $3 < h$ for Newman's approximations for the Struve functions.

6. Conclusion

Although the approximations (4) may arguably be regarded as a trivial modification of the exact expressions (3) for the wave components $W(h, v)$ and $W_h(h, v)$, this modification is useful for practical applications because it yields global approximations to W and W_h . Indeed, expressions (3)–(4) for W and W_h only involve elementary

functions of real arguments and the intrinsic Fortran Bessel functions $J_0(h)$ and $J_1(h)$.

These global analytical approximations to the wave components $W(h, v)$ and $W_h(h, v)$ and the global analytical approximations to the local-flow components $L(h, v)$ and $L_h(h, v)$ given in [1,2] provide particularly simple approximations to G and ∇G that are valid for $0 \leq h$ and $v \leq 0$, i.e. within the entire flow region. These global approximations can be evaluated simply and efficiently, and are especially well suited for parallel computations because they avoid the 'if statements' that are required if different approximations are used in complementary contiguous subregions of the flow region $0 \leq h$ and $v \leq 0$.

The illustrative numerical applications given in [2] and here provide solid evidence that the global approximations to the Green function G and its gradient are sufficiently accurate for practical applications. Notably, the contribution of the singular terms in the local-flow components L and ∇L can be evaluated accurately because these singular terms are explicitly apparent in (2) and in the analytical approximations for L and L_h given in [1,2].

Acknowledgments

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